

Improving Network Measurement Efficiency Through Multiadaptive Sampling

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Abstract. Sampling techniques play a key role in achieving efficient network measurements by reducing the amount of traffic processed while trying to maintain the accuracy of network statistical behavior estimation.

Despite the evolution of current techniques regarding the correctness of network parameters estimation, the overhead associated with the volume of data involved in the sampling process is still considerable. In this context, this paper proposes a new technique for multiadaptive traffic sampling based on linear prediction, which allows to reduce significantly the traffic under analysis, keeping the representativeness of samples in capturing network behavior.

A proof-of-concept, evaluating this technique for real traffic traces representing distinct traffic profiles, demonstrates the effectiveness of the proposal, outperforming classic techniques both in accuracy and data volumes processed.

1 Introduction

Traffic sampling techniques are extensively used to reduce the impact of performing traffic measurements on operational equipment. Their main objective is to select a subset of packets which will be used to estimate network parameters correctly, avoiding processing all network traffic. In adaptive sampling techniques the packet selection process may change dynamically during the measurement period, based on the value of a measured reference parameter. Despite the evolution on adaptive techniques correctness, their main target is not focused on reducing the overhead associated with the volume of data involved in the sampling process. This aspect directly impacts on monitoring costs and efficiency.

In this context, this paper presents a new multiadaptive traffic sampling technique based on linear prediction, which aims to reduce the volume of data handled in the network measurements without compromising estimation accuracy. For this purpose, the traffic selection process considers the levels of network activity, being configured to reduce measurement impact when the network activity increases and the measurement process tends to be heavier for measurement points. The multiadaptive behavior of this proposal is achieved considering both the sampling interval and the sample size as adaptive parameters, bounded by thresholds that guarantee the representativeness of samples in capturing network behavior.

The remaining of this document is organized as follows: related work is discussed in Section 2; the multiadaptive sampling technique is described in Section 3; the evaluation results are presented in Section 4; finally, the conclusions are drawn in Section 5.

2 Related Work

Common sampling adaptive techniques are usually based on Fuzzy Logic and Linear Prediction. In fuzzy logic techniques [2] [6], a controller adjusts the sampling rate based on past similar experiences, determining the most appropriate action for current traffic conditions [3]. This approach requires a long-term database to store the knowledge and the possible action for each situation.

Linear prediction based techniques [4] [5] try to forecast network behavior based on an observed parameter in past samples. In these techniques, when the prediction is correct, the sampling rate can be reduced, while inaccurate predictions indicate a change in network activity and, therefore, an increase in sampling rate is required to determine the new pattern behavior [3]. However, if the sampling frequency increases more resources will be required from the measurement point, precisely at the most critical moment of its operation.

The sampling technique proposed in this paper decreases resource consumption related to the processing, storage and transmission of captured packets in high network activity periods, maintaining the accuracy of network statistical behavior estimation.

3 Multiadaptive Sampling Technique

The multiadaptive sampling technique proposed in this work, grounded by previous research [3] [1], improves the sampling reactivity and efficiency, considering both the interval between samples and the sample size as adjustable parameters.

For this, the technique takes into account the last N samples to estimate the future value of the reference parameter. Thus, using linear prediction, the expected value Xp of the reference parameter for the next collected sample is defined by Equation 1 [3],

$$Xp = X[N] + \frac{\Delta T_{current}}{N-1} \sum_{i=1}^{N-1} \left(\left| \frac{X[i+1] - X[i]}{\Delta T[i]} \right| \right) \quad (1)$$

where $X[N]$ is the most recente sample and ΔT is the time between the end of the sample $X[i]$ and the beginning of the sample $X[i+1]$.

When a new sample is collected, the corresponding value of the reference parameter S is compared with the expected value Xp . Based on a factor of change $m = \left| \frac{Xp - X[N]}{S - X[N]} \right|$ a set of rules is applied to define ΔT_{next} and ΔS_{next} used to schedule the start and the size of the next sample. The factor m returns a result close to 1 when the expected value Xp is close to the current value of S . The range of values that satisfies this condition is defined as $m_{min} < 1 < m_{max}$. If $m < 0.9$ the reference parameter is changing faster than predicted. This behavior indicates more network activity than expected. On the other hand, if $m > 1.1$ the value of the reference parameter is changing slower than predicted. However, if the value of S does not change since the last sample $X[N]$, m assumes an undefined value, which indicates that the network is stable. These values are set experimentally due to its good performance for multiple traffic types [3].

Table 1 lists the rules used to generate ΔT_{next} and ΔS_{next} from the current value of m . These rules reduce the amount of packets captured in high network activity periods.

Table 1. Rules to define the next interval between samples and sample size

current m	ΔT_{next}	ΔS_{next}
$m < m_{min}$	$\Delta T_{next} = m * \Delta T_{current}$	$\Delta S_{next} = \Delta S_{current} - (\Delta S_{current}/k), k = 4$
$m_{min} \leq m \leq m_{max}$	$\Delta T_{next} = \Delta T_{current}$	$\Delta S_{next} = \Delta S_{current}$
$m_{max} < m$	$\Delta T_{next} = \Delta T_{current} + 1seg$	$\Delta S_{next} = \Delta S_{current} + (\Delta S_{current}/k), k = 10$
$m_{undefined}$	$\Delta T_{next} = 2 * \Delta T_{current}$	$\Delta S_{next} = \Delta S_{current} + (\Delta S_{current}/k), k = 10$

An additional constraint limits the interval between samples and the sample size, guaranteeing a minimum number of samples to new predictions and avoiding to capture all traffic, respectively. These limits, tuned experimentally, define the minimum interval between samples as 0.1s and the maximum as 8s. The minimum sample size is defined as 0.1s and the maximum as 2s.

4 Evaluation Results

The proof-of-concept of the multiadaptive sampling strategy aims to determine its ability to capture the network behavior correctly with reduced overhead. Therefore, a statistical comparison of the reference parameter (network throughput with order of prediction $N = 3$) obtained through traffic sampling and the total traffic is carried out. In addition, the performance of the proposed strategy is compared to other commonly used techniques, the Systematic Time-based - ST technique (RFC 5475) and the Adaptive Linear Prediction - LP technique [3]. A set of statistical parameters is evaluated for four different traffic types, aiming at demonstrating the effectiveness of the proposal in distinct traffic scenarios. The *SIGCOMM 2008* scenario corresponds to a four hour capture of IEEE 802.11a communications (CRAWDAD). The *OC-48* scenario corresponds to a capture in a backbone link OC - Optical Carrier 48 (CAIDA). The *VoIP-SIP* represents around twenty hours of VoIP traffic, using SIP and G.711 encoding. The *Video Streaming* scenario corresponds to the capture of a live stream transmission in High Definition 720p encoded in MPEG-4 using RTP as transport protocol.

Table 2 shows that the multiadaptive technique (MA) leads to a significant overhead reduction for all traffic types under consideration, reducing up to 95% the data amount involved in the traffic characterization. The *data volume* corresponds to the sum, in Mbytes, of all packets collected with each sampling technique. The equivalence between the total traffic behavior and the behavior resulting from the sampling process is determined using *throughput* (kbps), *coefficient of variation* (CV), *correlation* and *relative mean error* (RME). CV measures the variability of time series, helping to characterize the burstiness of traffic. Correlation measures the statistical relationship between the sampled traffic and the total traffic. RME is used to assess the discrepancy between the mean of the total traffic and its sampled version [1].

From these results it is possible to verify that, despite the significant reduction on traffic volume considered, the multiadaptive technique ability to capture the real traffic behavior correctly is not compromised. Moreover, for several statistical parameters, the proposed technique outperforms the conventional techniques, which is even more significant when considering the high decrease on measurement overhead.

Table 2. Overall estimated behavior

Traffic/Parameter	Total	ST	LP	MA	Traffic/Parameter	Total	ST	LP	MA
SIGCOMM08					SIP VoIP				
Data volume (MB)	2382.92	395.97	511.93	120.49	Data volume (MB)	168.68	27.41	32.27	9.03
Throughput (kbps)	730.95	728.76	1130.27	803.65	Throughput (kbps)	21.85	21.30	50.49	22.03
RME		0.002	0.54	0.09	RME		0.02	1.31	0.008
CV	1.90	2.07	1.56	1.93	CV	5.28	5.41	2.90	5.08
Correlation		0.91	0.91	0.89	Correlation		0.71	0.88	0.86
OC-48					Video streaming				
Data volume (MB)	3189.67	533.39	454.92	165.72	Data volume (MB)	101.76	17.02	14.86	4.85
Throughput (kbps)	87103.35	87390.26	86467.86	86825.99	Throughput (kbps)	1275.84	1279.17	2998.53	1425.09
RME		0.003	0.007	0.003	RME		0.002	1.35	0.11
CV	0.10	0.09	0.10	0.13	CV	3.93	3.96	2.50	3.66
Correlation		0.84	0.81	0.82	Correlation		0.99	0.98	0.98

5 Conclusions

This paper has presented a multiadaptive traffic sampling technique based on linear prediction. The multiadaptive functionality is associated with the ability to change the interval between samples and the sample size according to the network observed activity. Controlling these two variables and defining appropriate bounds for their values, this technique reduces significantly the overhead associated with the traffic sampling process while maintaining the accuracy in estimating the statistical network behavior when taking throughput as reference parameter. Future work will be centred in further developing the technique regarding simultaneous estimation of parameters such as packet loss, delay and jitter, both in real time and in large-scale testbeds.

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